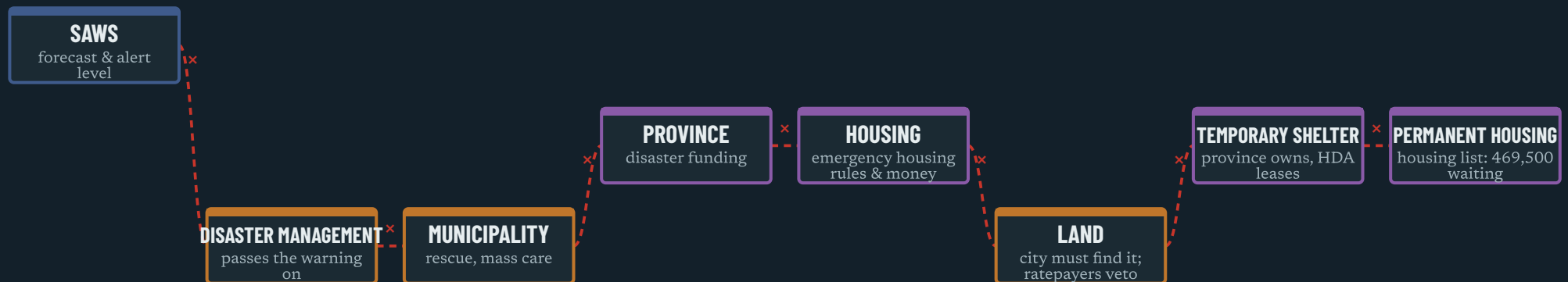


WHO OWNS THE FLOOD?

This is South Africa's chain from flood forecast to permanent home: 8 links, held by 3 spheres of government. Nobody owns it from end to end. The next 9 pages show how we found that out.



8 LINKS · 3 SPHERES · 7 HAND-OFFS · 0 END-TO-END OWNERS

Chain reconstructed from the National Housing Code Vol. 4 (Emergency Housing Programme), eThekweni's briefing to Parliament (8 Mar 2023), the Auditor-General's flood-relief report (Aug 2022) and National Treasury disaster grants 2022/23. Details on page 8.

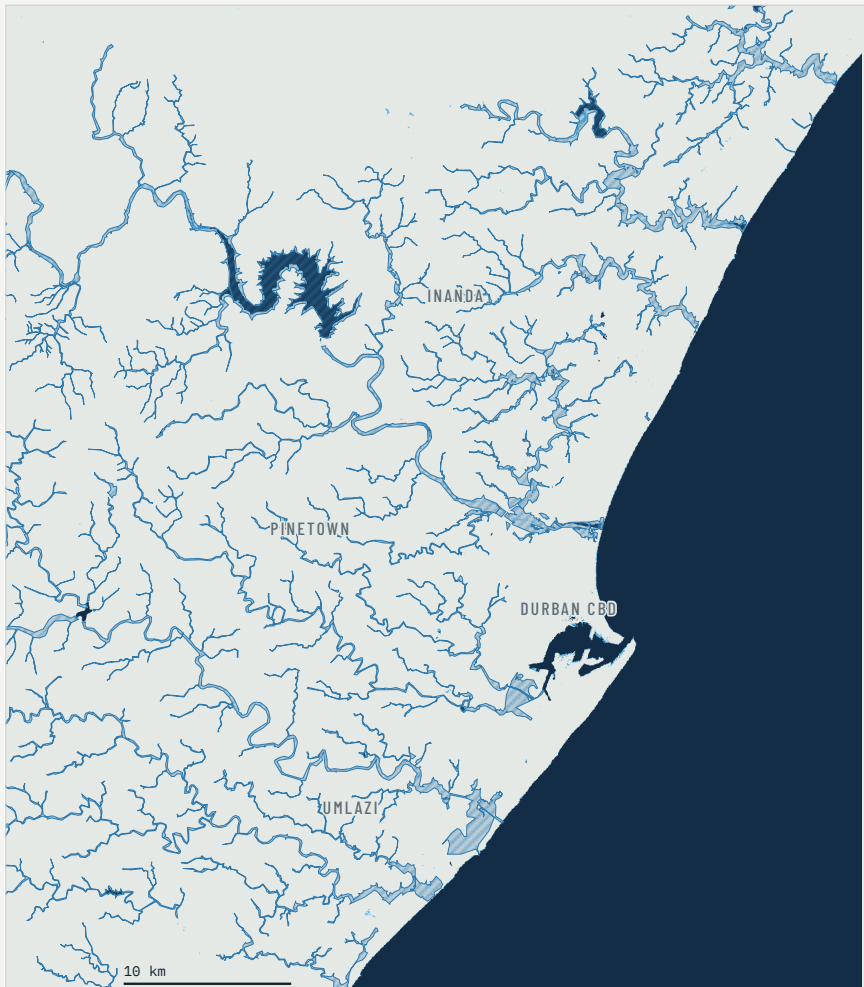
WHERE DOES THE WATER ACTUALLY GO?

DURBAN, SEEN FROM SPACE

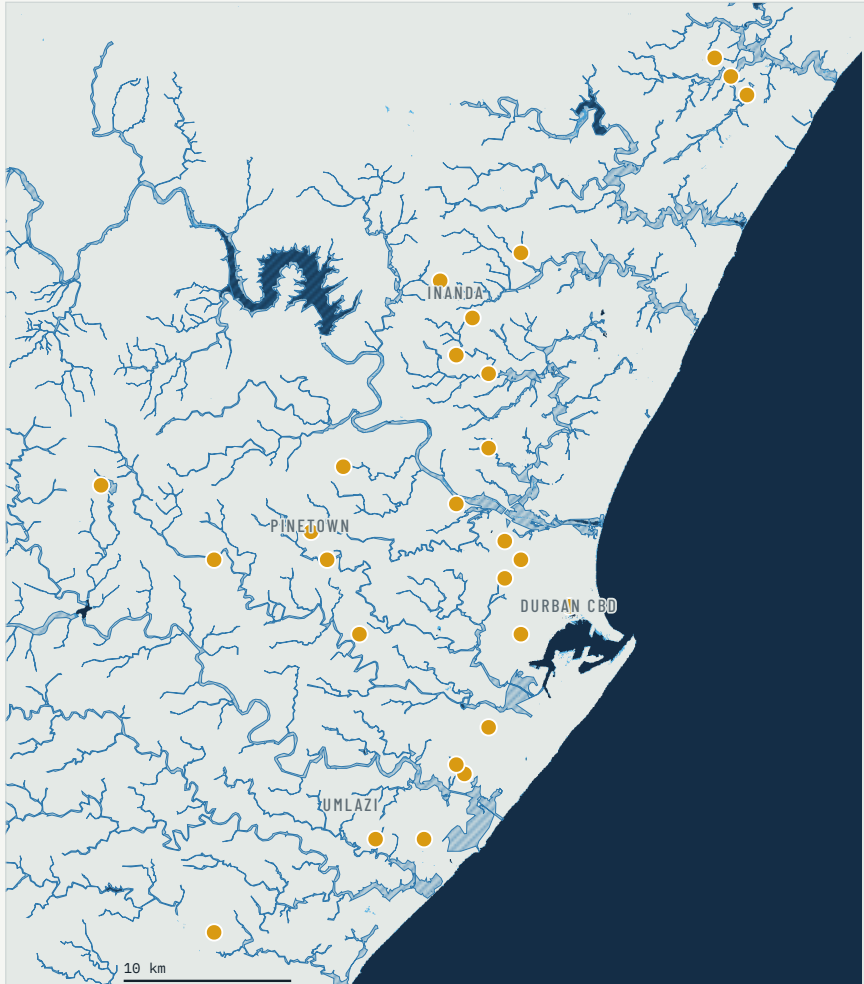
Bright blue is ground that Landsat satellites saw under water now and then between 1984 and 2021: ground that floods and dries. **Dark blue** is permanent water.

Hatched is eThekweni's own 1-in-100-year floodplain. The city knows where the water goes, and it publishes the map.

■ intermittent water ■ permanent water ■ 100-year floodplain



Sources: EC JRC Global Surface Water, occurrence 1984–2021 (Pekel et al. 2016, *Nature* 540:418–422), global-surface-water.appspot.com · eThekweni Municipality, "Flood Plain 100yr", [gis-ethekweni.opendata.arcgis.com](https://gis-ethekweni.opendata.arcgis.com/dataset/e33fb655723e49fab434a23a8ca3d1d2_0) (dataset e33fb655723e49fab434a23a8ca3d1d2_0).



WHERE DID WE PUT PEOPLE?

28 CAMPS ON THE CITY'S MAP

Amber marks transit camps and emergency accommodation, 2022–2026.

11 of 28

Camps inside the city's floodplain or within 250 m of it (39%). **Random ground across Durban: 34%.** Within 1 km: 22 of 28 camps, against 84% of random ground.

The flood map made no difference to where people were put.

Nationally, the list has 84 "temporary" sites. The oldest active one, the Langa containers in Cape Town, has lasted 24 years.

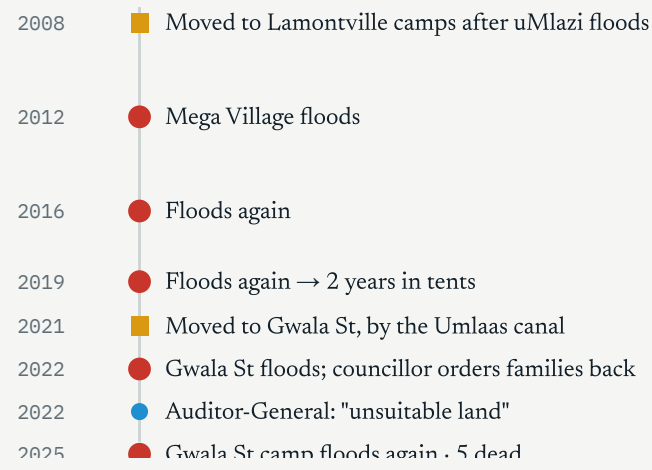
Sources: "South Africa TRAs 2022–2026" site list (84 sites, each with its own sources), data/tra_sites_2022_2026.csv · Spatial test against the eThekweni floodplain, with 300 seeded random points across Durban's urban area as the baseline: methods/scripts/build_tra_exposure.ps1 · Most coordinates are accurate only to $\pm 1\text{--}3$ km, so read each site as indicative.



WHAT HAPPENS WHEN THE PEOPLE WE MOVED ARE FLOODED AGAIN?

THE WATER COMES BACK

3 Durban camps and 7 sites nationally have documented flooding after people moved in (plus 2 low-confidence reports). Only 30 of the 84 sites have been searched one by one, so these are minimum counts. The Auditor-General flagged unsuitable siting in August 2022. In February 2025, 5 people died at the Gwala Street camp in Lamontville.



Sources: Lamontville: IOL 26 Feb 2025; Amnesty International, *Flooded and Forgotten*, AFR 53/0369/2025 · Gwala St: Mail & Guardian 9 May 2022; GroundUp, "We have been moved from one place to another"; Daily Maverick 23 Mar 2022 · Isipingo: IOL; Rising Sun · Auditor-General, flood relief special report, Aug 2022 · Umbilo move: site list #54, Sunday Tribune Oct 2025 · Reflooding sweep: data/raw/tra_reflood_sweep.json (Madamfana: GroundUp, Sep 2023; Denver: Sowetan, Jan 2024; Kampies: GroundUp 2020 and Daily Maverick, May 2026; Motherwell NU30: GroundUp, May 2023; Blikkiesdorp: GroundUp, Mar 2024; Feb 2025 camp named as Gwala St by IOL, medium confidence)

IS THIS UNIQUELY SOUTH AFRICAN? NO.

In each case below, people were moved by the state and the new site then flooded.

Lamontville & Gwala St

South Africa · eThekweni

Isipingo camp

South Africa · eThekweni

Perumbakkam

India · Chennai

Semmencheri

India · Chennai

Kasiglahan Village

Philippines · Rizal

Corail-Cesselesse

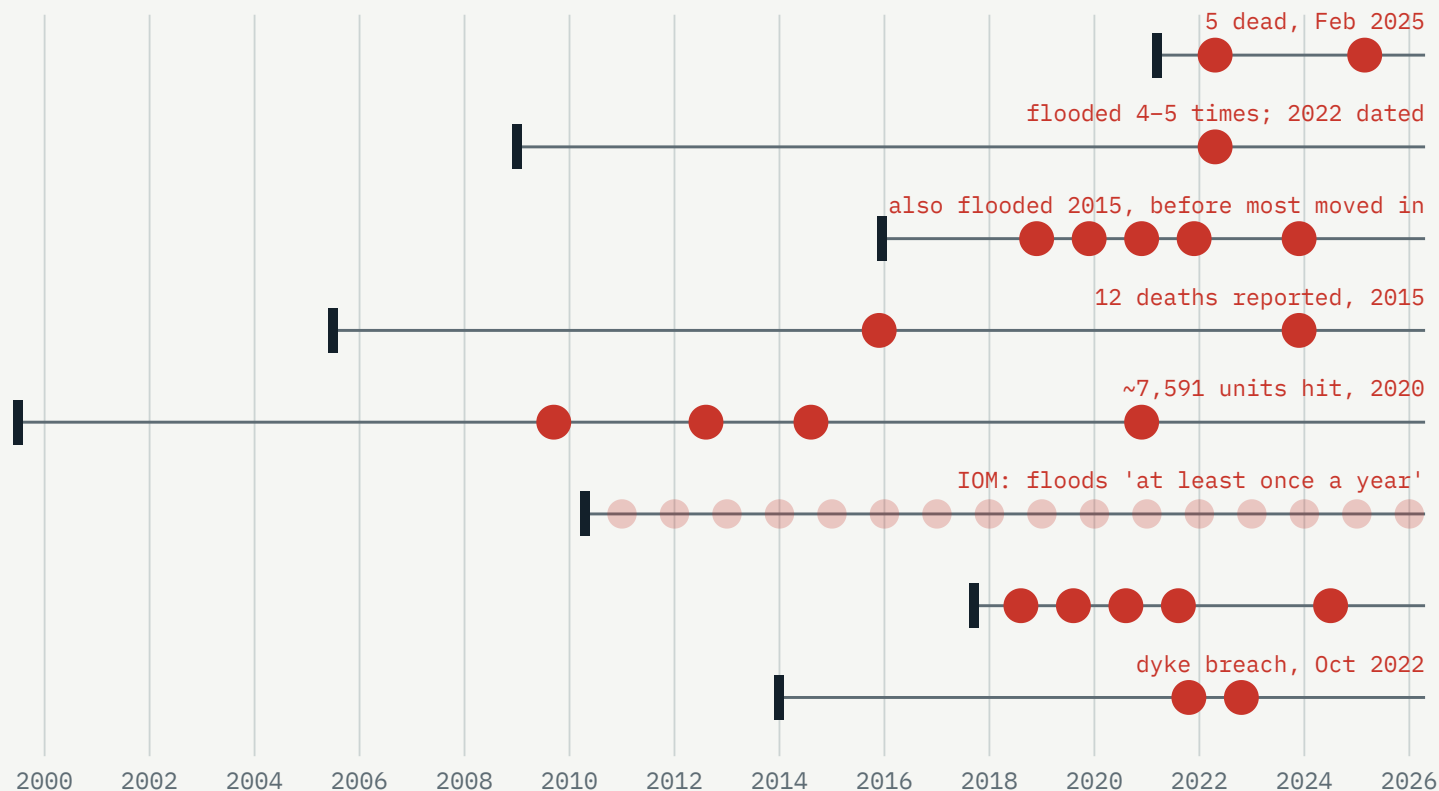
Haiti

Cox's Bazar camps

Bangladesh

Bentiu camp

South Sudan



Sources: Perumbakkam: IRCDOC flood audit; The News Minute · Semmencheri: Coelho, *Caravan* · Kasiglahan: Philstar; Lauer et al. 2026, *Progress in Disaster Science* · Corail: Haiti Grassroots Watch, citing IOM (July 2010) · Cox's Bazar: HRW 2019; UN Bangladesh 2024 · Bentiu: Eye Radio; UN South Sudan · Full case files: data/raw/relocation_into_floodzones_*.json, relocation_deep_does.json.

SOME COUNTRIES MOVE PEOPLE AND KEEP THEM DRY. SO WHAT IS DIFFERENT?

Grantham

Australia · hilltop land swap

Vunidogoloa

Fiji · village's own clan land

Iwanuma

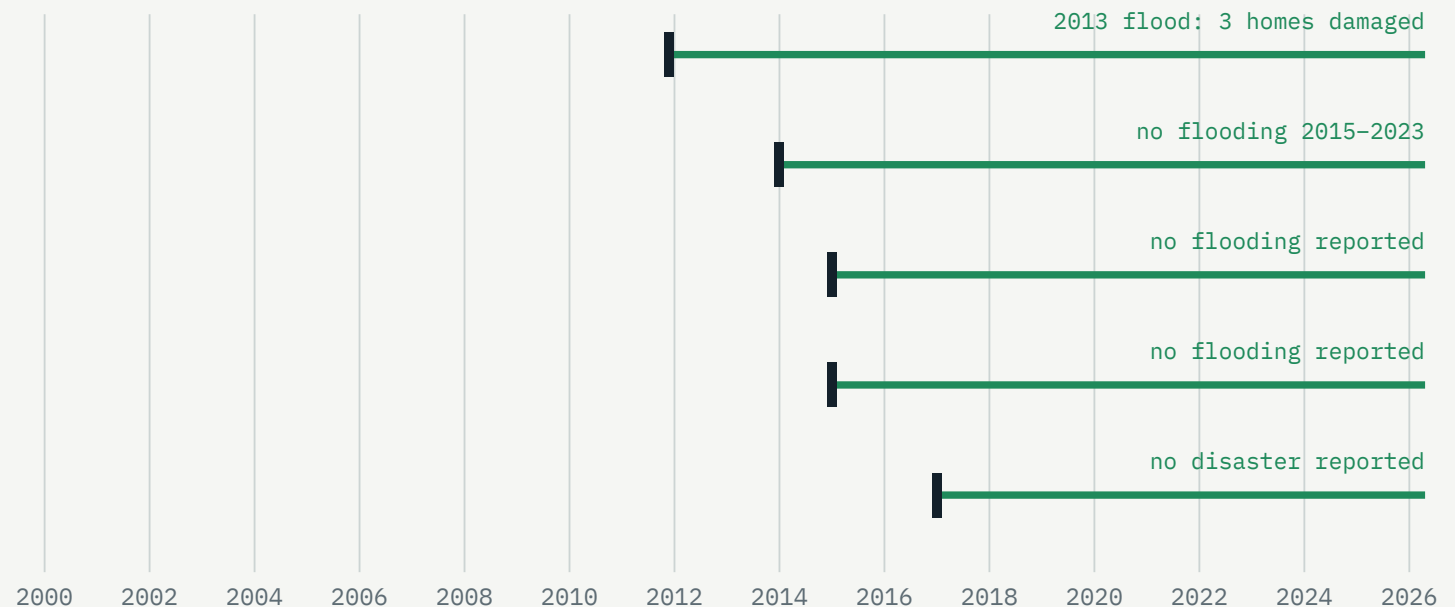
Japan · resident committee, 28 meetings

Overdiepse polder

Netherlands · farms on 6 m mounds

Gramalote

Colombia · renters became owners



None of these is perfect. Grantham helped only property owners, Gramalote took six years, and Vunidogoloa's drainage was never finished. None of the new sites has yet faced a flood as big as the one that displaced its people.

Sources: Grantham: Okada et al. 2014, *IJDRR* · Vunidogoloa: McMichael et al. 2019, *Geographical Journal*; 2021 (PMC8072796) · Iwanuma: Japan Reconstruction Agency case study · Overdiepse: Roth & Winnubst 2015 · Gramalote: Natural Hazards Center quick-response report; Miraflores resettlement plan (2013) · data/raw/relocation_deep_doesnt.json.

IS IT THAT THE STATE DOESN'T CARE? THAT PEOPLE ARE POOR?

THOSE MAKE IT WORSE. THEY DON'T EXPLAIN IT.

~~THE STATE DOESN'T CARE~~

The state acted fast: an R516m first round of flood funding, 71 transit camps built, and an Auditor-General audit within 4 months.

PLENTY OF ACTIVITY; THE GAP IS ELSEWHERE

ReliefWeb/Treasury 2022 · IOL 2021 · AGSA 2022

~~THE PEOPLE ARE POOR~~

Vunidogoloa, a village of 26 households, moved to safety. Gramalote turned its 40% renters into owners. Poor communities were moved safely where someone owned the outcome.

RAISES EXPOSURE; DOESN'T DECIDE IT

McMichael 2019 · Natural Hazards Center

~~CORRUPTION~~

Found in 3 of the 4 failures where there was evidence to check. But Mozambique and Somalia failed without it.

MAKES IT WORSE; DOESN'T CAUSE IT

IOL R32m probe 2023 · AP (Haiti) · rootcause matrix

~~INCOMPETENCE~~

In every failure, someone had flagged the site first: the Auditor-General, a UN agency, a national audit, a study.

THEY KNEW

AGSA 2022 · IOM 2010 · CAG 2014 · 2004 study

~~CLIMATE~~

Grantham flooded again in 2013. The families it had moved to the hilltop stayed dry.

THE SAME RAIN FALLS ON EVERY CASE

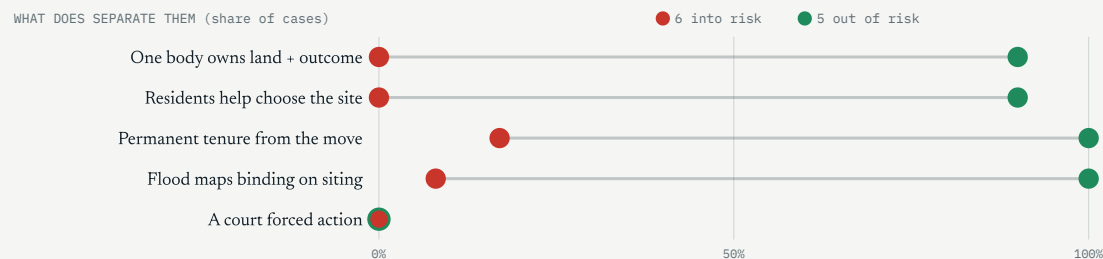
Okada et al. 2014

~~MONEY~~

Haiti had large amounts of international money and still built on flood land. Fiji and Colombia got it right.

DOESN'T SEPARATE THE CASES

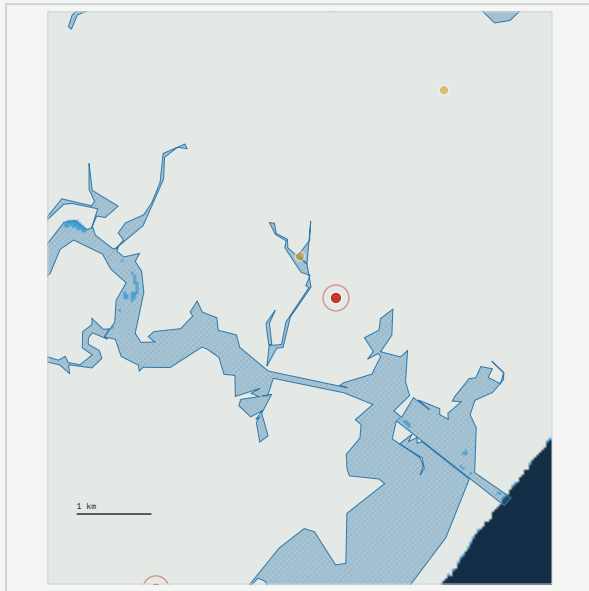
Haiti Grassroots Watch · McMichael 2021



Sources: 11 cases scored on 10 factors from the sourced findings: data/relocation_rootcause_matrix.csv, built by methods/scripts/build_relocation_rootcause.ps1 · The cases were chosen because of how they turned out, so read this as a pattern, not a statistical test · Consistent with Bower et al. 2023, *Nature Climate Change* 13:919-926 (community-driven relocations did better).

WHAT DOES "BINDING" MEAN?

A MAP ON FILE VS A MAP THAT CAN SAY NO



Gwala St camp, Lamontville (red), with the city's own 1-in-100-year floodplain hatched. The site list's coordinate is approximate ($\pm 1-3$ km); reports place the camp beside the Umlaas canal.

39% **34%**

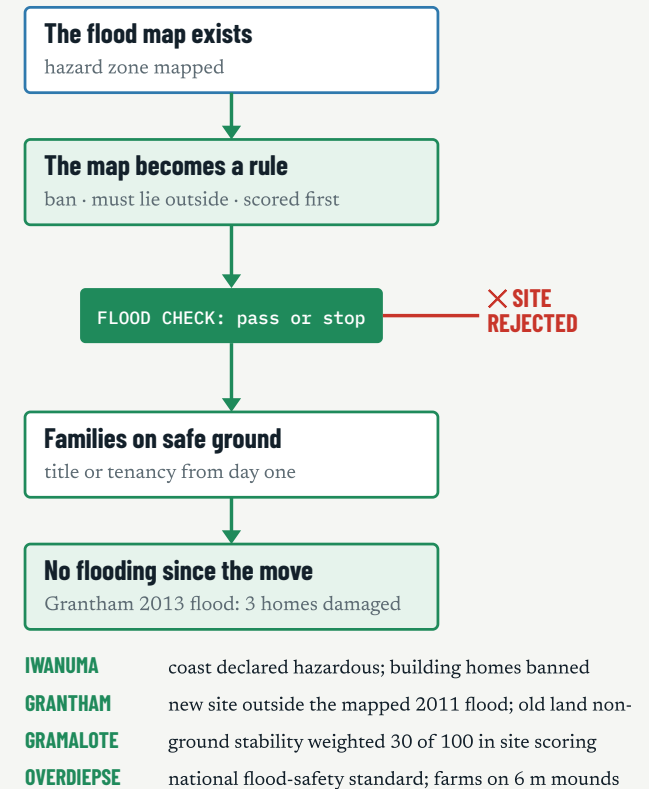
Camps vs **random ground**: the share inside or within 250 m of the floodplain. The map made no difference.

Sources: eThekweni Flood Plain 100yr · National Housing Code 2009, Vol. 4, Emergency Housing Programme (temporary assistance; permanent-structure norms "will not apply"; fast-tracked or exempt planning and EIA procedures) · Auditor-General flood-relief report, Aug 2022 ("some units... built on unsuitable land") · Mail & Guardian 9 May 2022 · IOL / Amnesty 2025 · Iwanuma: Japan Reconstruction Agency; Grantham: Okada et al. 2014; Gramalote: Miraflores resettlement plan 2013 (stability weighted 30/100); Overdiepse: Roth & Winnubst 2015 · Floodplain test: methods/scripts/build_tra_exposure.ps1.

DURBAN · NOT BINDING

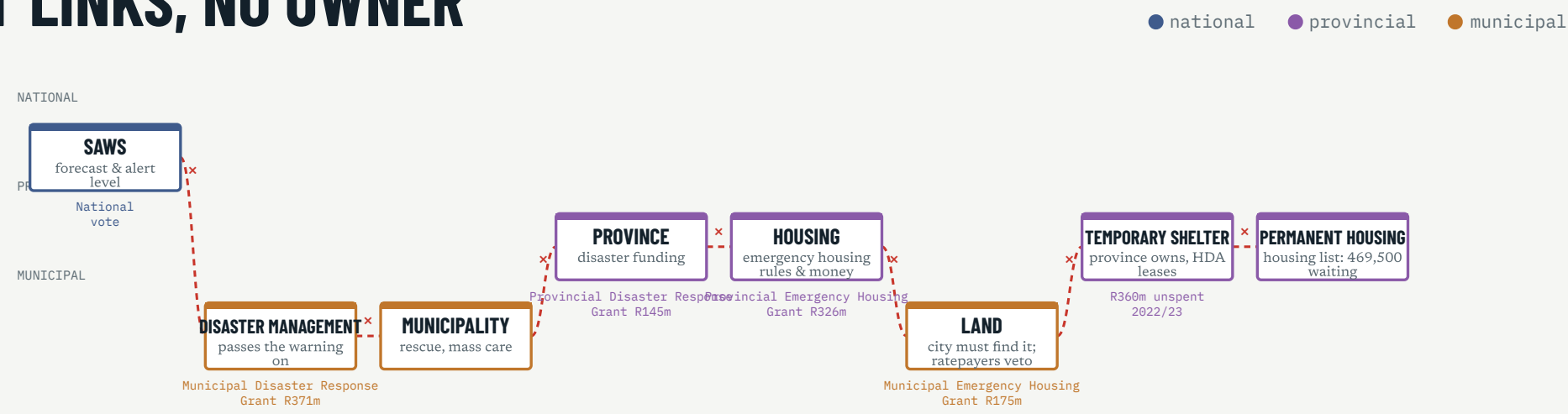


SUCCESS CASES · BINDING



DIFFERENT INSTITUTIONS. DIFFERENT BUDGETS. DIFFERENT AUTHORITY.

EIGHT LINKS, NO OWNER



8 LINKS · 3 SPHERES · 7 HAND-OFFS · 0 END-TO-END OWNERS

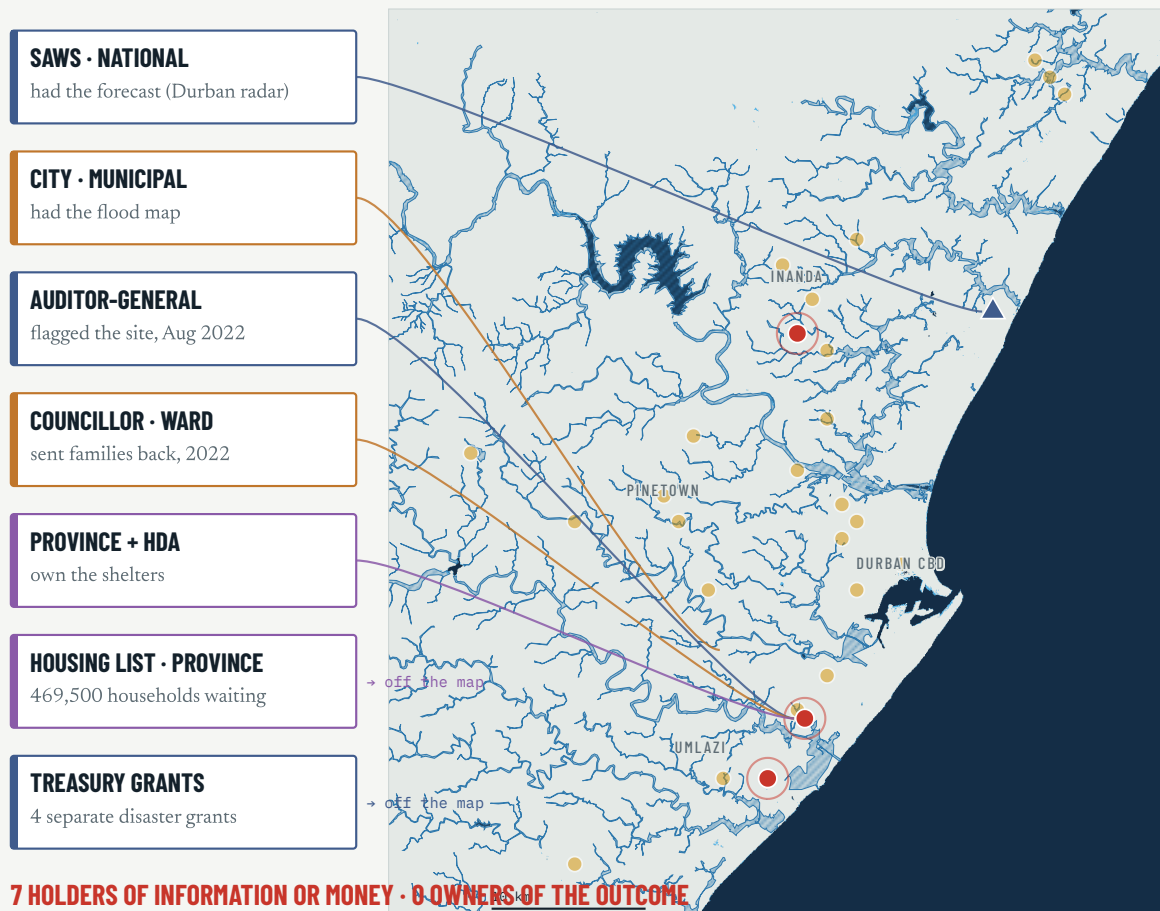
Displaced families: lease, no title, no seat at any link

GRANTHAM · IWANUMA · GRAMALOTE · OVERDIEPSE · VUNIDOGOLOA



SAME 8 LINKS · 1 OWNER · RESIDENTS AT THE TABLE

Sources: National Housing Code 2009, Vol. 4, Emergency Housing Programme · eThekweni Metro briefing to Parliament, 8 Mar 2023 (land ownership; constraints: "coordination across line departments and spheres") · 2022/23 disaster grants: National Treasury (Municipal Disaster Response R371m; Provincial Disaster Response R145m; Provincial Emergency Housing R326m; Municipal Emergency Housing R175m) · GroundUp: R360m unspent, 0 of 2,224 units built · Housing backlog 469,500 households (eThekweni) · Success models: Okada 2014; Reconstruction Agency; Oliver-Smith & Arenas; Roth & Winnubst 2015.



THE PROBLEM WASN'T THAT NOBODY KNEW.

THE PROBLEM WAS THAT NOBODY OWNED THE WHOLE OUTCOME.

The forecast sat with SAWS. The flood map sat with the city. The warning about the site sat with the Auditor-General. The shelters belonged to the province. **The families sat on the riverbank.**

SOUTH AFRICA HAS THE INFORMATION. THE INSTITUTIONS THAT POSSESS IT DO NOT CONTROL THE DECISIONS THAT DETERMINE EXPOSURE.

Sources: SAWS Durban S-band radar location: WMO Radar Database / OSCAR (data/observing_network.json) · eThekweni Flood Plain 100yr · Auditor-General, Aug 2022 ("some units... built on unsuitable land") · Mail & Guardian, 9 May 2022 (councillor ordered families back to the flood-prone camp) · IOL / Amnesty 2025 (Lamontville deaths) · Housing Code Vol. 4 (province owns the temporary units).

Question. Why does South Africa move flood-displaced people onto land that floods again, and what do the countries that don't do this have in common?

Sites. "South Africa TRAs 2022–2026": 84 temporary relocation sites (TRAs, TEAs, TRUs and transit camps), each with its own sources. Transcribed to `data/tra_sites_2022_2026.csv`. Why each site was created is coded from the list's type and notes; 78 sites have coordinates, mostly estimates (± 1 –3 km).

Satellite water. JRC Global Surface Water occurrence 1984–2021, 30 m Landsat (Pekel et al. 2016). Tiles cached in `data/raw/gsw_tiles`; Durban mosaic recoloured by `methods/scripts/recolur_durban_water.ps1` (occurrence under 50% = intermittent).

Floodplain test. eThekwinini "Flood Plain 100yr" FeatureServer. Each Durban site is tested as inside, within 250 m, or within 1 km. Baseline: 300 seeded random points across Durban's urban area, on the land side of the coast (`data/raw/durban_random_baseline.csv`).

Reflooding. Counts only floods documented at the relocation site after people moved in: the site list's status notes, our case research, and a sweep of 30 of the 84 sites (`data/raw/tra_reflood_sweep.json`; high and medium confidence counted, low reported separately). If a site isn't counted, that means no flood was documented, not that it stayed dry.

World cases. 24 relocation-into-risk cases (`data/relocation_cases.csv`); in-depth reviews of 5 failures and 5 successes (`data/raw/relocation_deep_*.json`), scored on 10 factors (`data/relocation_rootcause_matrix.csv`).

Limits. The world cases were chosen because of how they turned out, so this is a pattern, not proof. Codes are our reading of sourced findings. Some quotes rest on search excerpts and are tagged in the JSON. Gramalote was a landslide, not a flood. The Beledweyne evidence is a submitted manuscript that may not yet be peer-reviewed.

Key references. Pekel et al. 2016, *Nature* · Bower et al. 2023, *Nature Climate Change* · Okada et al. 2014, *IJDRR* · Roth & Winnubst 2015 · McMichael et al. 2019 · Oliver-Smith & Arenas · Coelho, *Caravan* · Arabindoo 2016, *City* · Lauer et al. 2026 · Amnesty International 2025 (AFR 53/0369/2025) · Auditor-General of South Africa 2022 · National Housing Code Vol. 4 · eThekwinini briefing to Parliament, 8 Mar 2023.

Reproduce. Repository github.com/wandilemthiyane94-jpg/mde-dataviz: run `methods/scripts/build_tra_exposure.ps1`, then `build_relocation_rootcause.ps1`. Interactive version: prototype/story/moved-into-the-water.html.